

Lista 7

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Bibliotecas

```
library(DBI)
library(RSQLite)
```

1

```
path <- getwd()
fname <- file.path(path, "uwmadison.sqlite3")
conn <- dbConnect(RSQLite::SQLite(), fname)
```

2

```
dbGetQuery(conn, "
  SELECT DISTINCT instructors.name
  FROM subjects
  INNER JOIN subject_memberships ON subjects.code = subject_memberships.subject_code
  INNER JOIN sections ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
  INNER JOIN teachings ON sections.uuid = teachings.section_uuid
  INNER JOIN instructors ON teachings.instructor_id = instructors.id
  WHERE subjects.abbreviation = 'STAT'
  ORDER BY instructors.name
")
```

name
ABIGAIL BENZINE
ADIN-CRISTIAN ANDREI
AKICHIKA OZEKI
ALAN HUANG
ALBERTO DEL PIA
ALBRECHT KLEMM
ALEXANDER COVINGTON
ALEXANDER FISH
ALEXANDER KISELEV
ALYSSA DIGILIO
... 377 linhas no total

```
dbGetQuery(conn, "
  SELECT COUNT(DISTINCT teachings.instructor_id) AS total
  FROM subjects
  INNER JOIN subject_memberships ON subjects.code = subject_memberships.subject_code
  INNER JOIN sections ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
  INNER JOIN teachings ON sections.uuid = teachings.section_uuid
  WHERE subjects.abbreviation = 'STAT'
")
```

total
379

3 - professor mais difícil

```
dbGetQuery(conn, "
SELECT instructors.name,
AVG(
  (4.0 * grade_distributions.a_count +
   3.5 * grade_distributions.ab_count +
   3.0 * grade_distributions.b_count +
   2.5 * grade_distributions.bc_count +
   2.0 * grade_distributions.c_count +
   1.0 * grade_distributions.d_count) /
  (grade_distributions.a_count + grade_distributions.ab_count +
   grade_distributions.b_count + grade_distributions.bc_count +
   grade_distributions.c_count + grade_distributions.d_count +
   grade_distributions.f_count)
) AS GPA
FROM grade_distributions
INNER JOIN sections ON grade_distributions.course_offering_uuid = sections.course_offering_uuid
AND grade_distributions.section_number = sections.number
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
INNER JOIN subject_memberships ON grade_distributions.course_offering_uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
AND (grade_distributions.a_count + grade_distributions.ab_count +
     grade_distributions.b_count + grade_distributions.bc_count +
     grade_distributions.c_count + grade_distributions.d_count +
     grade_distributions.f_count) > 0
GROUP BY instructors.id, instructors.name
ORDER BY GPA
LIMIT 1
")
```

name	GPA
JAMES D KUELBS	2.6375

3 - professor mais fácil

```
dbGetQuery(conn, "
SELECT instructors.name,
AVG(
  (4.0 * grade_distributions.a_count +
   3.5 * grade_distributions.ab_count +
   3.0 * grade_distributions.b_count +
   2.5 * grade_distributions.bc_count +
   2.0 * grade_distributions.c_count +
   1.0 * grade_distributions.d_count) /
  (grade_distributions.a_count + grade_distributions.ab_count +
   grade_distributions.b_count + grade_distributions.bc_count +
   grade_distributions.c_count + grade_distributions.d_count +
   grade_distributions.f_count)
) AS GPA
FROM grade_distributions
INNER JOIN sections ON grade_distributions.course_offering_uuid = sections.course_offering_uuid
AND grade_distributions.section_number = sections.number
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
INNER JOIN subject_memberships ON grade_distributions.course_offering_uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
AND (grade_distributions.a_count + grade_distributions.ab_count +
     grade_distributions.b_count + grade_distributions.bc_count +
     grade_distributions.c_count + grade_distributions.d_count +
     grade_distributions.f_count) > 0
GROUP BY instructors.id, instructors.name
ORDER BY GPA DESC
LIMIT 1
")
```

name	GPA
SRIKANTHMADHAVAN ARAVAMUTHAN	4

3 - disciplina mais dificil

```
dbGetQuery(conn, "
SELECT courses.name,
AVG(
  (4.0 * grade_distributions.a_count +
   3.5 * grade_distributions.ab_count +
   3.0 * grade_distributions.b_count +
   2.5 * grade_distributions.bc_count +
   2.0 * grade_distributions.c_count +
   1.0 * grade_distributions.d_count) /
  (grade_distributions.a_count + grade_distributions.ab_count +
   grade_distributions.b_count + grade_distributions.bc_count +
   grade_distributions.c_count + grade_distributions.d_count +
   grade_distributions.f_count)
) AS GPA
FROM grade_distributions
INNER JOIN course_offerings ON grade_distributions.course_offering_uuid = course_offerings.uuid
INNER JOIN courses ON course_offerings.course_uuid = courses.uuid
INNER JOIN subject_memberships ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
AND (grade_distributions.a_count + grade_distributions.ab_count +
     grade_distributions.b_count + grade_distributions.bc_count +
     grade_distributions.c_count + grade_distributions.d_count +
     grade_distributions.f_count) > 0
GROUP BY courses.uuid, courses.name
ORDER BY GPA
LIMIT 1
")
```

name	GPA
Introduction to the Theory of Probability	2.891568

3 - disciplina mais facil

```
dbGetQuery(conn, "
SELECT courses.name,
AVG(
  (4.0 * grade_distributions.a_count +
   3.5 * grade_distributions.ab_count +
   3.0 * grade_distributions.b_count +
   2.5 * grade_distributions.bc_count +
   2.0 * grade_distributions.c_count +
   1.0 * grade_distributions.d_count) /
  (grade_distributions.a_count + grade_distributions.ab_count +
   grade_distributions.b_count + grade_distributions.bc_count +
   grade_distributions.c_count + grade_distributions.d_count +
   grade_distributions.f_count)
) AS GPA
FROM grade_distributions
INNER JOIN course_offerings ON grade_distributions.course_offering_uuid = course_offerings.uuid
INNER JOIN courses ON course_offerings.course_uuid = courses.uuid
INNER JOIN subject_memberships ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
AND (grade_distributions.a_count + grade_distributions.ab_count +
     grade_distributions.b_count + grade_distributions.bc_count +
     grade_distributions.c_count + grade_distributions.d_count +
     grade_distributions.f_count) > 0
GROUP BY courses.uuid, courses.name
ORDER BY GPA DESC
LIMIT 1
")
```

name	GPA
Nonparametric Statistics and Machine Learning Methods	4

4

```
dbDisconnect(conn)

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```